

SUPPLEMENTARY MATERIAL FOR AN APPLICATION NOTE

Supplementary material for: Return-aware validation of an earnings coefficient under household masking in the Current Population Survey

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Supplementary material

This supplement states the restricted theoretical result used to interpret focal estimability, reports every prespecified R3 and R4 design cell, and records the numerical protocol and provenance. Probability statements refer only to the controlled laws stated below.

Appendix S1. Restricted theory and scope

S1.1. Finite contrasts and likelihood limits

For observed logistic design W_R , put $S = \text{diag}(2y_R - 1)W_R$ and $\mathcal{C}_R = \{v : Sv \geq 0\}$. Rows with strictly positive signed margin along a recession direction become perfectly fitted in that limit. Let $\mathcal{F}_R$ be the rows remaining after taking the union of such rows over the cone. The limiting likelihood depends on the parameter through $W_{\mathcal{F}_R}\theta$; this is the recession-geometry construction described by Geyer (2009).

Suppose the limiting model has a finite maximizer. Its equivalent parameter representations differ by elements of $\ker(W_{\mathcal{F}_R})$. Consequently $c^\top \theta$ is the same in all these representations if and only if $c \in \text{row}(W_{\mathcal{F}_R})$: this follows from the orthogonality of the row and null spaces. A path with $c^\top v \neq 0$ in the recession cone also prevents a finite contrast along that path. The implemented procedure checks the entire cone and the remaining row space. Orthogonality to one chosen direction does not suffice. These statements concern the likelihood limit and identification; they do not provide a standard error under random household masking.

S1.2. A profile model with sparse events

The following result supplies a mask-law argument for a restricted model. For each H , fix all records, including their outcomes, and partition them into a bulk set B_H and a fixed number q of disjoint strata $S_{j,H}$. This partition is specified before masking. Households may cross the sets. Write $x_i \in \mathbb{R}^d$, with fixed d , and use the working predictors $x_i^\top b$ in the bulk and $x_i^\top b + \alpha_j$ in stratum j . In this subsection b is the shared coefficient vector; $c^\top b$ is the focal contrast. All regressors are available in an observed household. The model defines a likelihood projection, without assuming a stochastic model for the fixed outcomes.

For independent $R_h \sim \text{Bernoulli}(p)$, $0 < p < 1$ fixed, let

$$\ell_R^P(b) = \ell_{B,R}(b) + \sum_{j=1}^q \sup_{\alpha_j \in \mathbb{R}} \ell_{j,R}(b, \alpha_j). \quad (\text{S1.1})$$

An empty stratum contributes zero. A stratum containing only zeros or only ones also has supremum zero, at intercept $-\infty$ or $+\infty$, respectively; its contribution is constant in b . Let $b_{F,H}$ and $\hat{b}_{E,H}$ maximize the full-record and masked profile objectives. The target is $c^\top b_{F,H}$. Let $b_{0,H}$ and $\hat{b}_{B,H}$ denote the corresponding bulk-only roots, used only in the proof.

The assumptions concern a sequence of complete record arrays.

- (P1) Each household has at most b_* records, $0 < c_N \leq N/H \leq C_N$, and $\max_i \|x_i\| \leq C$, for constants independent of H .

- (P2) The full bulk root $b_{0,H}$ lies in a fixed compact set. With $\mu_i(b) = \text{expit}(x_i^\top b)$ and $J_{B,H}(b) = \sum_{i \in B_H} x_i x_i^\top \mu_i(b) \{1 - \mu_i(b)\}$, $A_H = J_{B,H}(b_{0,H})/H \rightarrow A$, where A is positive definite.
- (P3) The total rare-stratum event count $M_H = \sum_j \sum_{i \in S_{j,H}} y_i$ is $o(\sqrt{H})$. The number of rare-stratum records may be proportional to H .
- (P4) For $u_{h,H} = \sum_{i \in h \cap B_H} x_i \{y_i - \mu_i(b_{0,H})\}$,

$$\Omega_H = \frac{1-p}{pH} \sum_h u_{h,H} u_{h,H}^\top \longrightarrow \Omega, \quad c^\top A^{-1} \Omega A^{-1} c > 0.$$

P2 does not require the normalized full information for (b, α) to be nonsingular. A numerical implementation additionally needs focal optimization error $o_P(H^{-1/2})$ relative to the mathematical optimizer for the conclusions below to transfer to its output.

Lemma S1.1 (Uniform profile bounds). *For a stratum with m observed events, its profile score $r_j(b)$ and negative profile Hessian $J_j^P(b)$ satisfy*

$$\|r_j(b)\| \leq 2Cm, \quad \|J_j^P(b)\| \leq 4C^2m. \quad (\text{S1.2})$$

The bounds hold for every mask and every finite b , taking both quantities to be zero for empty or single-outcome strata.

Proof. If $0 < m < n$, the unique profiled intercept solves $\sum_i \mu_i = m$. Thus $r_j = \sum_i x_i(y_i - \mu_i)$ and $\sum_i |y_i - \mu_i| = 2(m - \sum_{y_i=1} \mu_i) \leq 2m$. Writing $w_i = \mu_i(1 - \mu_i)$ and $\bar{x}_j = \sum_i w_i x_i / \sum_i w_i$, implicit differentiation gives $J_j^P = \sum_i w_i (x_i - \bar{x}_j)(x_i - \bar{x}_j)^\top$. Use $\|x_i - \bar{x}_j\| \leq 2C$ and $\sum_i w_i \leq m$. In a boundary case the profile objective is constant, proving the stated convention and bounds there as well. $\square$

Proposition S1.2 (Restricted focal mask limit). *Under P1–P4, $b_{F,H}$ exists uniquely for all sufficiently large H , and a unique finite $\hat{b}_{E,H}$ exists with probability tending to one. Moreover,*

$$\begin{aligned} b_{F,H} - b_{0,H} &= O(M_H/H), \\ \hat{b}_{E,H} - \hat{b}_{B,H} &= O_P(M_H/H). \end{aligned} \quad (\text{S1.3})$$

$$\hat{b}_{E,H} - b_{F,H} = A_H^{-1} \frac{1}{H} \sum_h (R_h/p - 1) u_{h,H} + o_P(H^{-1/2}). \quad (\text{S1.4})$$

Consequently $\sqrt{H} c^\top (\hat{b}_{E,H} - b_{F,H})$ converges in distribution to a centered normal variable with variance $c^\top A^{-1} \Omega A^{-1} c$.

Proof. At $b_{0,H}$ the masked bulk score is $\sum_h (R_h - p) u_{h,H} = O_P(\sqrt{H})$, since $\sum_h u_{h,H} = 0$. P1 bounds the score and Hessian contributions of each household. The normalized masked Hessian differs uniformly from $pJ_{B,H}(b)/H$ by $o_P(1)$ on a fixed small neighborhood of $b_{0,H}$. To see uniformity, apply the variance bound $O(H^{-1})$ to entries on a finite covering net and use bounded logistic third derivatives between net points. Compactness handles the moving centers. P2 therefore gives curvature at least aH on this neighborhood, with probability tending to one, for a fixed $a > 0$.

For displacement t from $b_{0,H}$ the outward radial derivative of the masked bulk objective is at most $\|t\| \|U_{B,R}(b_{0,H})\| - aH\|t\|^2$. On a sphere of radius $K/\sqrt{H}$ it is inward with arbitrarily high probability for sufficiently large K . Concavity gives an interior root and $\hat{b}_{B,H} - b_{0,H} = O_P(H^{-1/2})$. Positive local information implies that the observed bulk design has full column rank. Its positive logistic weights then make the bulk objective strictly concave at every finite argument, so this root is the unique global maximizer.

Lemma S1.1 bounds the total profile-score perturbation by $2CM_H = o(\sqrt{H})$ uniformly over masks. Adding it preserves the inward derivative argument and strict concavity. Comparing the two zero scores along the segment between the roots gives $\|\hat{b}_{E,H} - \hat{b}_{B,H}\| \leq 2CM_H/(aH)$ on the regular event. The deterministic full-record argument is the same. This proves (S1.3), including masks on which stratum intercepts separate. No derivative through a selected face is taken.

A bulk score expansion and the uniform Hessian approximation give $\hat{b}_{B,H} - b_{0,H} = (pA_H)^{-1}H^{-1} \sum_h (R_h - p)u_{h,H} + o_P(H^{-1/2})$. Both perturbations in (S1.3) are $o_P(H^{-1/2})$, which transfers this expansion to the full profile target and proves (S1.4). Its leading household terms are independent and centered. P1 bounds each unnormalized summand, and P4 gives positive limiting focal variance. The scalar triangular-array Lindeberg condition follows because each bounded summand divided by $\sqrt{H}$ tends uniformly to zero. The central limit theorem and Slutsky's lemma prove the conclusion. $\square$

For a distributional statement, the statistic can be assigned a fixed value on the event where no optimizer exists, whose probability tends to zero. This convention is not used to fill failed experimental intervals.

S1.3. Observed-data variance for the restricted model

At the observed profile fits, set

$$\begin{aligned}\hat{J}_R &= J_{B,R}(\hat{b}_E) + \sum_j \hat{J}_{j,R}^P, \\ \hat{e}_h &= \sum_{i \in h \cap B_H} x_i(y_i - \hat{\mu}_i) + \sum_j \sum_{i \in h \cap S_{j,H}} (x_i - \hat{x}_j)(y_i - \hat{\mu}_i), \quad R_h = 1, \\ \hat{V}_E &= (1-p)\hat{J}_R^{-1} \left(\sum_{h: R_h=1} \hat{e}_h \hat{e}_h^\top \right) \hat{J}_R^{-1}.\end{aligned}\tag{S1.5}$$

Boundary-stratum contributions are zero. All contributions in a household are added before taking the outer product, including households crossing the bulk and strata. Under P1–P4, $H\hat{V}_E \rightarrow_P A^{-1}\Omega A^{-1}$.

For completeness, the bounded-household law of large numbers gives

$$H^{-1} \sum_h R_h u_h u_h^\top - pH^{-1} \sum_h u_h u_h^\top \rightarrow_P 0.$$

Replacing $b_{0,H}$ by $\hat{b}_E$ changes this by $o_P(1)$ because their difference is $O_P(H^{-1/2})$ and the score derivatives are bounded. The bulk bread divided by H tends to pA . The extra profile bread is at most $4C^2M_H = o(H)$ in norm. Further, $\sum_{j,i} \|(x_i - \hat{x}_j)(y_i - \mu_i)\| \leq$

$4CM_H$ on observed rare rows. Their household sums remain uniformly bounded by P1, so adding them changes the meat, including bulk cross terms, by $O(M_H) = o(H)$. Combining these limits proves the variance claim.

Studentization therefore gives asymptotic mask coverage for the focal profile interval. The factor $H/(H-1)$ used in the experiment does not change this limit. There is no additional $1/p$ in (S1.5): the bread is an unweighted observed sum, asymptotic to pHA . This proves consistency for the CLT covariance; convergence of actual finite- H second moments requires separate tail control. A fixed solver tolerance alone does not establish the required vanishing optimization error.

S1.4. An explicit sequence and the limits of transfer

Take one record per household. Each of t bulk blocks contains four records with $z = -1$, one of them an event, and four with $z = 1$, three of them events. Set $x = (1, z)^\top$. Add a rare stratum of $4t$ records, equally split over the two z values, with two events, one at each value. Then $H = 12t$, $b_0 = (0, \log 3)^\top$, $A_H = I_2/8$ and $H^{-1} \sum_h u_h u_h^\top = I_2/8$. P1–P4 hold although the rare record fraction is $1/3$. Its full-record intercept is finite at every t and tends to $-\infty$. Hiding both rare event households while observing a rare nonevent produces separation, with probability

$$(1-p)^2 \{1 - (1-p)^{4t-2}\} \longrightarrow (1-p)^2 > 0. \quad (\text{S1.6})$$

The focal slope nevertheless has CLT variance $8(1-p)/(pH)$. At $p = 0.25$ the separation probability tends to 0.5625 and the variance multiplied by H tends to 24. Deterministic derivative, matrix and sequence checks of this example are computational checks, not a coverage simulation.

The information-growth condition matters. If only two households have nonzero z , one containing the bulk block above and the other its reversed outcomes, adding balanced households with $z = 0$ does not increase slope information. Observing only the first, only the second, both, or neither gives slope $\log 3$, $-\log 3$, 0, or an unidentified slope, with probabilities $p(1-p)$, $p(1-p)$, p^2 , and $(1-p)^2$. Finite returned slopes need not become consistent. Also, inverse likelihood information is not a mask variance: in the balanced block it lacks the factor $1-p$ even before clustering is considered.

The CPS contains overlapping adjustment groups and hinge terms; their scores cannot be bounded using a separate free stratum-intercept equation without further argument. One CPS month cannot certify P1–P4 along a sequence. The synthetic experiment has a predefined rare stratum and bounded household size, but only one value of H and unbounded normal regressors. It is not a verification of P1–P4. The proposition also contains no fitted imputation learner or CV step. It therefore does not establish general CPS random-face coverage or coverage for the mean, Gaussian or augmented procedures.

S1.5. Reference displacement and coverage identities

For a fixed array, let $G_F(\theta) = N^{-1} \sum_i s_i(\theta)$ and $G_D(\theta) = N^{-1} \sum_i \{ps_i(\theta) + (1-p)t_i^F(\theta)\}$. Assume finite roots θ_F, θ_D , continuous differentiability along their joining segment, and invertibility of $\bar{A}_D = -\int_0^1 DG_D\{\theta_F + t(\theta_D - \theta_F)\} dt$. Integration along

that segment and $G_F(\theta_F) = 0$ give

$$\theta_D - \theta_F = \bar{A}_D^{-1} \frac{1}{N} \sum_i (1-p) \{t_i^F(\theta_F) - s_i(\theta_F)\}. \quad (\text{S1.7})$$

This finite-root identity relates score discrepancy to reference displacement. It is not a formula for the expectation of a refitted finite-mask estimator.

An additional, conditional argument explains possible coverage changes. Suppose interval return tends to one, $\sqrt{H}(\hat{\gamma} - \gamma_D)/\sigma_H \Rightarrow N(0, 1)$, $\sqrt{H}\hat{s}/\sigma_H \rightarrow_P \kappa > 0$, and $\sqrt{H}(\gamma_D - \gamma_F)/\sigma_H \rightarrow d$, with σ_H bounded above and away from zero. For $z = z_{1-\alpha/2}$, Slutsky's lemma gives

$$P\{\gamma_F \in [\hat{\gamma} - z\hat{s}, \hat{\gamma} + z\hat{s}]\} \longrightarrow \Phi(z\kappa - d) - \Phi(-z\kappa - d). \quad (\text{S1.8})$$

Here the CLT is assumed, not established for each implemented procedure. If return does not tend to one, its joint law with the statistic is needed. In all cases, for return event I and coverage among returns C , $P(I \cap C) = P(I)P(C \mid I) \leq P(I)$ when $P(I) > 0$. These identities motivate the distinct reported denominators; they do not impute coverage to unavailable intervals.

Appendix S2. Complete fixed-array results

Tables S2.1–S2.6 preserve the fixed-array targets, all 54 method cells, both prespecified paired contrasts, all 15 representative masks and every non-available status. The notation “joint” counts a non-returned interval as not covering; “conditional” restricts the denominator to returned intervals. The Clopper–Pearson intervals for the individual binary rates and the 4,000-draw mask bootstrap intervals for continuous and paired summaries were computed from the same planned mask indices. The tables report the point summaries; paired intervals are retained where they bear directly on comparisons.

Table S2.1.: Complete-data projection and sensitivity targets for every fixed monthly array.

Month	Specification	N	H	β_F	OR	ΔP (pp)
2025-01	Main adjustment	7,085	4,801	-0.1019	0.932	-0.385
2025-01	Hours excluded	7,085	4,801	-0.2713	0.829	-1.104
2025-01	Outgoing weights	7,085	4,801	-0.0742	0.950	-0.255
2025-07	Main adjustment	6,390	4,254	-0.1228	0.918	-0.457
2025-07	Hours excluded	6,390	4,254	-0.2487	0.842	-0.973
2025-07	Outgoing weights	6,390	4,254	-0.0584	0.960	-0.199
2026-01	Main adjustment	6,344	4,332	-0.0025	0.998	-0.009
2026-01	Hours excluded	6,344	4,332	-0.1557	0.898	-0.602
2026-01	Outgoing weights	6,344	4,332	-0.0175	0.988	-0.059

Table S2.2.: All fixed-array method cells. Return, joint coverage and conditional coverage are percentages; bias, RMSE and length use coefficient units.

Month	p	Method	M	Ret.	Joint	Cond.	Bias	RMSE	Length	SE/SD
2025-01	0.25	CC	6,400	89.0	84.7	95.2	-0.009	0.114	0.436	0.995
2025-01	0.25	Mean	6,400	100.0	88.4	88.4	-0.136	0.280	0.895	0.949
2025-01	0.25	Gaussian	6,400	100.0	93.5	93.5	-0.009	0.111	0.405	0.940
2025-01	0.25	Aug.	6,400	100.0	95.5	95.5	-0.006	0.109	0.424	1.009
2025-01	0.25	Ext. CC	6,400	100.0	95.0	95.0	-0.010	0.114	0.437	0.992
2025-01	0.25	Firth	6,400	100.0	94.3	94.3	-0.018	0.110	0.417	0.998
2025-01	0.50	CC	6,400	99.8	94.3	94.5	-0.001	0.063	0.240	0.973
2025-01	0.50	Mean	6,400	100.0	86.3	86.3	-0.061	0.113	0.355	0.964
2025-01	0.50	Gaussian	6,400	100.0	93.8	93.8	-0.002	0.062	0.233	0.961
2025-01	0.50	Aug.	6,400	100.0	94.7	94.7	0.001	0.059	0.227	0.989
2025-01	0.50	Ext. CC	6,400	100.0	94.5	94.5	-0.001	0.063	0.240	0.973
2025-01	0.50	Firth	6,400	100.0	94.4	94.4	-0.006	0.062	0.234	0.975
2025-01	0.75	CC	2,500	100.0	94.9	94.9	0.000	0.035	0.135	0.995
2025-01	0.75	Mean	2,500	100.0	89.5	89.5	-0.024	0.047	0.155	0.988
2025-01	0.75	Gaussian	2,500	100.0	94.8	94.8	-0.001	0.034	0.130	0.988
2025-01	0.75	Aug.	2,500	100.0	95.0	95.0	0.000	0.033	0.127	0.999
2025-01	0.75	Ext. CC	2,500	100.0	94.9	94.9	0.000	0.035	0.135	0.995
2025-01	0.75	Firth	2,500	100.0	94.4	94.4	-0.004	0.034	0.133	0.996
2025-07	0.25	CC	2,500	79.8	72.4	90.7	0.021	0.179	0.632	0.930
2025-07	0.25	Mean	2,500	100.0	92.4	92.4	-0.093	0.392	1.316	0.903
2025-07	0.25	Gaussian	2,500	100.0	89.2	89.2	0.013	0.171	0.604	0.917
2025-07	0.25	Aug.	2,500	100.0	91.6	91.6	0.021	0.171	0.633	0.969
2025-07	0.25	Ext. CC	2,500	100.0	90.8	90.8	0.020	0.179	0.635	0.934
2025-07	0.25	Firth	2,500	100.0	89.6	89.6	0.010	0.172	0.606	0.930
2025-07	0.50	CC	2,500	99.8	91.8	92.0	0.008	0.100	0.372	0.957
2025-07	0.50	Mean	2,500	100.0	88.4	88.4	-0.058	0.163	0.555	0.944
2025-07	0.50	Gaussian	2,500	100.0	92.2	92.2	0.004	0.100	0.374	0.961
2025-07	0.50	Aug.	2,500	100.0	93.1	93.1	0.007	0.095	0.361	0.979
2025-07	0.50	Ext. CC	2,500	100.0	92.0	92.0	0.008	0.100	0.372	0.957
2025-07	0.50	Firth	2,500	100.0	91.6	91.6	0.001	0.099	0.365	0.954
2025-07	0.75	CC	2,500	100.0	93.8	93.8	0.002	0.058	0.216	0.958
2025-07	0.75	Mean	2,500	100.0	90.9	90.9	-0.026	0.072	0.252	0.964
2025-07	0.75	Gaussian	2,500	100.0	94.1	94.1	0.000	0.056	0.214	0.971
2025-07	0.75	Aug.	2,500	100.0	94.2	94.2	0.002	0.055	0.209	0.977
2025-07	0.75	Ext. CC	2,500	100.0	93.8	93.8	0.002	0.058	0.216	0.958
2025-07	0.75	Firth	2,500	100.0	93.4	93.4	-0.003	0.057	0.213	0.956
2026-01	0.25	CC	2,500	68.4	65.1	95.1	0.009	0.148	0.587	1.028
2026-01	0.25	Mean	2,500	100.0	93.0	93.0	0.008	0.329	1.250	0.982
2026-01	0.25	Gaussian	2,500	100.0	93.6	93.6	0.000	0.135	0.511	0.976
2026-01	0.25	Aug.	2,500	100.0	94.4	94.4	0.006	0.147	0.594	1.041
2026-01	0.25	Ext. CC	2,500	100.0	95.0	95.0	0.008	0.149	0.592	1.026
2026-01	0.25	Firth	2,500	100.0	94.2	94.2	-0.003	0.147	0.583	1.031
2026-01	0.50	CC	2,500	96.4	90.4	93.8	0.003	0.088	0.333	0.968
2026-01	0.50	Mean	2,500	100.0	92.9	92.9	0.000	0.140	0.533	0.973
2026-01	0.50	Gaussian	2,500	100.0	93.9	93.9	-0.001	0.082	0.312	0.970
2026-01	0.50	Aug.	2,500	100.0	94.4	94.4	0.002	0.083	0.324	1.001
2026-01	0.50	Ext. CC	2,500	100.0	93.8	93.8	0.003	0.088	0.332	0.970
2026-01	0.50	Firth	2,500	100.0	93.6	93.6	-0.004	0.088	0.333	0.972
2026-01	0.75	CC	2,500	100.0	94.3	94.4	0.001	0.050	0.191	0.979
2026-01	0.75	Mean	2,500	100.0	94.2	94.2	0.000	0.060	0.231	0.990
2026-01	0.75	Gaussian	2,500	100.0	94.7	94.7	0.000	0.047	0.182	0.990
2026-01	0.75	Aug.	2,500	100.0	94.7	94.7	0.001	0.047	0.184	1.000
2026-01	0.75	Ext. CC	2,500	100.0	94.4	94.4	0.001	0.050	0.191	0.979
2026-01	0.75	Firth	2,500	100.0	94.0	94.0	-0.003	0.050	0.192	0.980

Table S2.3.: Reference diagnostics for the Mean and Gaussian procedures. The last two columns use the common complete-data projection and the method-specific reference, respectively.

Month	p	Method	β_F	β_D	Mean $\hat{\beta}$	Joint _F	Joint _D
2025-01	0.25	Mean	-0.102	-0.267	-0.238	88.4	95.9
2025-01	0.25	Gaussian	-0.102	-0.108	-0.111	93.5	93.4
2025-01	0.50	Mean	-0.102	-0.171	-0.163	86.3	94.5
2025-01	0.50	Gaussian	-0.102	-0.106	-0.104	93.8	93.8
2025-01	0.75	Mean	-0.102	-0.127	-0.126	89.5	95.0
2025-01	0.75	Gaussian	-0.102	-0.104	-0.103	94.8	94.8
2025-07	0.25	Mean	-0.123	-0.312	-0.215	92.4	92.2
2025-07	0.25	Gaussian	-0.123	-0.127	-0.110	89.2	89.2
2025-07	0.50	Mean	-0.123	-0.203	-0.181	88.4	93.0
2025-07	0.50	Gaussian	-0.123	-0.126	-0.119	92.2	92.1
2025-07	0.75	Mean	-0.123	-0.152	-0.149	90.9	94.2
2025-07	0.75	Gaussian	-0.123	-0.124	-0.123	94.1	94.0
2026-01	0.25	Mean	-0.003	-0.018	0.006	93.0	93.2
2026-01	0.25	Gaussian	-0.003	-0.006	-0.002	93.6	93.6
2026-01	0.50	Mean	-0.003	-0.009	-0.002	92.9	93.1
2026-01	0.50	Gaussian	-0.003	-0.005	-0.003	93.9	94.0
2026-01	0.75	Mean	-0.003	-0.005	-0.003	94.2	94.6
2026-01	0.75	Gaussian	-0.003	-0.004	-0.003	94.7	94.5

Table S2.4.: Paired fixed-array contrasts. Coverage and return differences and relative length changes are percentage points or percent, followed where applicable by 95% paired-bootstrap intervals.

Month	p	Contrast	N_{both}	$\Delta\text{Ret.}$	$\Delta\text{Joint [MC]}$	$\Delta\text{Length [MC]}$	Rel. $\Delta\text{Length [MC]}$
2025-01	0.25	Aug.-Ext.	6,400	0.0	0.4 [-0.1, 0.9]	-0.013 [-0.014, -0.012]	-3.0 [-3.3, -2.7]
2025-01	0.25	Firth-Ext.	6,400	0.0	-0.8 [-1.1, -0.4]	-0.020 [-0.020, -0.020]	-4.6 [-4.7, -4.5]
2025-01	0.50	Aug.-Ext.	6,400	0.0	0.1 [-0.4, 0.6]	-0.013 [-0.013, -0.012]	-5.3 [-5.5, -5.1]
2025-01	0.50	Firth-Ext.	6,400	0.0	-0.1 [-0.4, 0.2]	-0.006 [-0.006, -0.006]	-2.4 [-2.4, -2.3]
2025-01	0.75	Aug.-Ext.	2,500	0.0	0.1 [-0.6, 0.8]	-0.008 [-0.008, -0.008]	-6.0 [-6.2, -5.9]
2025-01	0.75	Firth-Ext.	2,500	0.0	-0.6 [-1.0, -0.1]	-0.002 [-0.002, -0.002]	-1.6 [-1.6, -1.6]
2025-07	0.25	Aug.-Ext.	2,500	0.0	0.8 [-0.2, 1.8]	-0.002 [-0.006, 0.001]	-0.4 [-1.0, 0.2]
2025-07	0.25	Firth-Ext.	2,500	0.0	-1.3 [-1.9, -0.7]	-0.029 [-0.030, -0.028]	-4.6 [-4.8, -4.4]
2025-07	0.50	Aug.-Ext.	2,500	0.0	1.1 [0.3, 1.8]	-0.011 [-0.012, -0.009]	-2.8 [-3.2, -2.5]
2025-07	0.50	Firth-Ext.	2,500	0.0	-0.4 [-0.8, 0.1]	-0.007 [-0.008, -0.007]	-2.0 [-2.1, -1.9]
2025-07	0.75	Aug.-Ext.	2,500	0.0	0.4 [-0.4, 1.1]	-0.007 [-0.008, -0.007]	-3.4 [-3.6, -3.2]
2025-07	0.75	Firth-Ext.	2,500	0.0	-0.4 [-1.0, 0.1]	-0.003 [-0.003, -0.003]	-1.3 [-1.3, -1.2]
2026-01	0.25	Aug.-Ext.	2,500	0.0	-0.6 [-1.4, 0.3]	0.002 [-0.001, 0.005]	0.4 [-0.1, 0.9]
2026-01	0.25	Firth-Ext.	2,500	0.0	-0.8 [-1.3, -0.3]	-0.009 [-0.009, -0.008]	-1.5 [-1.6, -1.3]
2026-01	0.50	Aug.-Ext.	2,500	0.0	0.6 [-0.2, 1.3]	-0.008 [-0.009, -0.008]	-2.5 [-2.8, -2.3]
2026-01	0.50	Firth-Ext.	2,500	0.0	-0.2 [-0.7, 0.2]	0.000 [0.000, 0.000]	0.1 [0.0, 0.1]
2026-01	0.75	Aug.-Ext.	2,500	0.0	0.4 [-0.4, 1.2]	-0.007 [-0.007, -0.007]	-3.6 [-3.8, -3.5]
2026-01	0.75	Firth-Ext.	2,500	0.0	-0.4 [-0.9, 0.0]	0.001 [0.001, 0.001]	0.4 [0.4, 0.4]

Table S2.5.: Prespecified January 2025 masks on which ordinary full-vector CC had no valid root but the earnings contrast was row-space estimable. Every named union matched the reconstructed removed-row set exactly; all Extended CC and Firth entries returned.

p	Rep.	Certified group(s)	Removed	Event	HH	Row err.	Ext. CC [CI]	Aug. Firth [CI]
0.25	3	asian only	112		90	4.8e-16	-0.330 [-0.570, -0.091]	-0.398 -0.320 [-0.537, -0.104]
0.25	20	hours above 50	60		78	7.2e-16	-0.164 [-0.412, 0.083]	-0.315 -0.165 [-0.412, 0.082]

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Table S2.5 (continued)

p	Rep.	Certified group(s)	Removed	Event HH	Row err.	Ext. CC [CI]	Aug. Firth [CI]
0.25	24	below high school	129	90	5.5e-16	-0.042 [-0.267, 0.184]	-0.044 -0.045 [-0.253, 0.163]
0.25	67	hours outside 1-99	64	88	3.7e-16	0.009 [-0.261, 0.280]	-0.076 -0.007 [-0.279, 0.265]
0.25	168	age above 65	104	92	1.1e-15	-0.292 [-0.491, -0.092]	-0.202 -0.296 [-0.490, -0.103]
0.25	282	class worker 1	72	77	2.1e-16	-0.198 [-0.389, -0.007]	-0.245 -0.201 [-0.374, -0.028]
0.25	358	below high school; hours above 50	207	82	8.9e-16	-0.026 [-0.314, 0.261]	-0.145 -0.028 [-0.308, 0.251]
0.25	1230	low education; hours outside 1-99	147	80	9.7e-16	0.192 [-0.124, 0.508]	0.070 0.165 [-0.136, 0.467]
0.25	1774	below high school; asian only	231	80	1.8e-15	-0.065 [-0.327, 0.198]	-0.179 -0.076 [-0.330, 0.179]
0.25	2601	age below 25	197	85	5.6e-16	-0.195 [-0.379, -0.011]	-0.148 -0.219 [-0.393, -0.045]
0.25	2957	class worker 2	105	88	8.0e-16	-0.288 [-0.518, -0.057]	-0.259 -0.277 [-0.488, -0.065]
0.25	3070	below high school; age above 65	216	91	8.3e-16	-0.197 [-0.444, 0.050]	-0.227 -0.197 [-0.430, 0.036]
0.25	4986	hours above 50; age above 65	176	76	4.5e-16	-0.217 [-0.484, 0.049]	-0.214 -0.207 [-0.461, 0.047]
0.25	5208	below high school; class worker 2	245	72	1.4e-15	-0.153 [-0.386, 0.080]	-0.259 -0.174 [-0.393, 0.046]
0.50	512	below high school	241	174	2.7e-16	0.042 [-0.097, 0.181]	-0.019 0.034 [-0.105, 0.173]

Table S2.6.: Non-available statuses in the fixed-array experiment. Cells and methods absent from this table had no non-available status.

Month	p	Method	Status	Count
2025-01	0.25	CC	no_valid_root	701
2025-01	0.50	CC	no_valid_root	14
2025-07	0.25	CC	no_valid_root	505
2025-07	0.50	CC	no_valid_root	6
2026-01	0.25	CC	no_valid_root	789
2026-01	0.50	CC	no_valid_root	91
2026-01	0.75	CC	no_valid_root	1

Appendix S3. Paired learner intervention

S3.1. Complete data-generating mechanism

Each repetition has $H = 800$ households with two records each. Independently draw $a_h, u_h, e_{hi}, v_{hi} \sim N(0, 1)$ and set $x_{hi} = 0.5a_h + \sqrt{0.75}e_{hi}$. Set $G_h = 1$ for the first 160 households and zero otherwise. In the bulk, independently conditional on x , draw $y_{hi} \sim \text{Bernoulli}\{\text{expit}(-1 + 0.3x_{hi})\}$. In the rare stratum, choose $m \in \{2, 12\}$ households uniformly without replacement; set the first member's outcome to one in those households and all remaining rare-stratum outcomes to zero. Define

$$z_{hi} = 0.4x_{hi} + 0.8y_{hi} + 0.2G_h + \delta(x_{hi}^2 - 1) + 0.6u_h + 0.8v_{hi}.$$

All normal draws are independent of the randomization used for outcomes, rare-event households and masking. The primary target solves the complete logistic score on $(1, z, x, G)$. It is recomputed for each array and need not equal a coefficient in the reverse conditional model that generated z . The base learner is $(1, x, G, y)$; its enriched version adds $x^2 - 1$. Thus the base conditional-mean family is correctly specified at $\delta = 0$, although its finite-sample ridge fit need not equal the true mean.

Household folds are $h \bmod 5$, for $h = 0, \dots, 799$. The generation stream uses `SeedSequence([20260916, 6, 194723, phase, rep, 0])` and the masking stream replaces the last component by 1; the production phase is 0. The same underlying stream is restarted for the four (m, δ) conditions within a repetition. One uniform per household supplies nested masks at $p = 0.25$ and 0.50 . Each B^+ fit shares the original array, target and mask. All learner and interval rules are those in Section S5.

S3.2. Complete results

The intervention reused the 1,000 archived C6 complete data sets, targets, household masks and fold assignments. It changed only the nuisance basis from $B = (1, x, G, y)$ to $B^+ = (1, x, G, y, x^2 - 1)$ for Mean, Gaussian and Augmented. No term was selected after observing the results. The complete-data method-specific Mean and Gaussian references were recomputed with B^+ . All 48 method-learner cells had 1,000 valid targets, point estimates and intervals; hence the archived failure table contains only `available=1,000` entries.

Table S3.1.: All C6 learner-intervention cells. Every target was valid and every interval returned in all 48 method-learner cells; coverage columns are percentages.

m	δ	p	Learner	Method	Ret.	Joint	Bias	RMSE	Length	Ref. shift	Ref. cov.
2	0	0.25	B	Mean	100.0	0.3	2.710	2.748	1.626	2.781	94.2
2	0	0.25	B	Gaussian	100.0	89.9	-0.031	0.132	0.452	-0.007	90.7
2	0	0.25	B	Aug.	100.0	95.0	0.008	0.127	0.499	0.000	95.0
2	0	0.25	B^+	Mean	100.0	0.4	2.668	2.708	1.654	2.774	94.7
2	0	0.25	B^+	Gaussian	100.0	89.6	-0.033	0.133	0.451	-0.008	90.9
2	0	0.25	B^+	Aug.	100.0	95.1	0.008	0.127	0.500	0.000	95.1
2	0	0.50	B	Mean	100.0	0.0	0.908	0.929	0.679	0.908	93.4
2	0	0.50	B	Gaussian	100.0	93.3	-0.009	0.072	0.273	-0.005	94.1
2	0	0.50	B	Aug.	100.0	94.8	0.003	0.071	0.285	0.000	94.8
2	0	0.50	B^+	Mean	100.0	0.0	0.903	0.924	0.677	0.906	93.1
2	0	0.50	B^+	Gaussian	100.0	93.7	-0.009	0.072	0.273	-0.005	93.7
2	0	0.50	B^+	Aug.	100.0	94.9	0.003	0.071	0.286	0.000	94.9
2	1	0.25	B	Mean	100.0	33.3	1.014	1.113	1.502	1.121	85.6
2	1	0.25	B	Gaussian	100.0	90.5	-0.005	0.069	0.239	0.009	90.4
2	1	0.25	B	Aug.	100.0	93.0	0.008	0.071	0.264	0.000	93.0
2	1	0.25	B^+	Mean	100.0	71.7	0.068	0.086	0.184	0.076	93.3
2	1	0.25	B^+	Gaussian	100.0	93.2	-0.008	0.038	0.137	-0.002	93.0
2	1	0.25	B^+	Aug.	100.0	95.0	0.001	0.036	0.146	0.000	95.0
2	1	0.50	B	Mean	100.0	10.3	0.279	0.300	0.350	0.285	93.5
2	1	0.50	B	Gaussian	100.0	93.3	0.002	0.039	0.147	0.005	92.9
2	1	0.50	B	Aug.	100.0	93.8	0.001	0.040	0.151	0.000	93.8
2	1	0.50	B^+	Mean	100.0	60.2	0.044	0.052	0.098	0.046	94.0
2	1	0.50	B^+	Gaussian	100.0	93.5	-0.003	0.021	0.081	-0.001	93.6
2	1	0.50	B^+	Aug.	100.0	94.7	0.000	0.021	0.084	0.000	94.7
12	0	0.25	B	Mean	100.0	0.1	2.584	2.622	1.548	2.660	94.0
12	0	0.25	B	Gaussian	100.0	89.8	-0.030	0.131	0.446	-0.007	90.6
12	0	0.25	B	Aug.	100.0	94.8	0.008	0.126	0.488	0.000	94.8
12	0	0.25	B^+	Mean	100.0	0.2	2.546	2.585	1.573	2.653	94.4

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Table S3.1 (continued)

m	δ	p	Learner	Method	Ret.	Joint	Bias	RMSE	Length	Ref. shift	Ref. cov.
12	0	0.25	B^+	Gaussian	100.0	89.8	-0.032	0.131	0.444	-0.008	90.5
12	0	0.25	B^+	Aug.	100.0	94.5	0.008	0.126	0.489	0.000	94.5
12	0	0.50	B	Mean	100.0	0.0	0.880	0.899	0.640	0.880	93.9
12	0	0.50	B	Gaussian	100.0	93.6	-0.008	0.071	0.268	-0.005	94.0
12	0	0.50	B	Aug.	100.0	95.2	0.003	0.070	0.279	0.000	95.2
12	0	0.50	B^+	Mean	100.0	0.0	0.875	0.895	0.639	0.878	93.6
12	0	0.50	B^+	Gaussian	100.0	93.9	-0.009	0.071	0.268	-0.005	94.0
12	0	0.50	B^+	Aug.	100.0	95.1	0.003	0.070	0.279	0.000	95.1
12	1	0.25	B	Mean	100.0	28.9	0.947	1.035	1.377	1.046	86.8
12	1	0.25	B	Gaussian	100.0	91.1	-0.001	0.068	0.235	0.011	90.3
12	1	0.25	B	Aug.	100.0	93.0	0.008	0.068	0.252	0.000	93.0
12	1	0.25	B^+	Mean	100.0	70.4	0.065	0.081	0.174	0.072	93.2
12	1	0.25	B^+	Gaussian	100.0	93.4	-0.008	0.036	0.131	-0.002	93.1
12	1	0.25	B^+	Aug.	100.0	95.3	0.001	0.035	0.140	0.000	95.3
12	1	0.50	B	Mean	100.0	9.1	0.266	0.284	0.326	0.271	93.2
12	1	0.50	B	Gaussian	100.0	93.1	0.004	0.039	0.143	0.007	92.7
12	1	0.50	B	Aug.	100.0	93.4	0.001	0.038	0.144	0.000	93.4
12	1	0.50	B^+	Mean	100.0	59.9	0.042	0.049	0.093	0.044	94.1
12	1	0.50	B^+	Gaussian	100.0	93.2	-0.003	0.020	0.078	-0.001	93.4
12	1	0.50	B^+	Aug.	100.0	94.6	0.000	0.020	0.080	0.000	94.6

Table S3.2.: Paired B^+ minus B learner effects for all C6 cells. Differences are followed by 95% repetition-bootstrap intervals; coverage and return differences are percentage points.

m	δ	p	Method	N_{both}	Δ Joint [MC]	Δ Ret. [MC]	Δ Length [MC]
2	0	0.25	Mean	1000	0.1 [0.0, 0.3]	0.0 [0.0, 0.0]	0.028 [0.024, 0.032]
2	0	0.25	Gaussian	1000	-0.3 [-1.0, 0.3]	0.0 [0.0, 0.0]	-0.002 [-0.002, -0.001]
2	0	0.25	Aug.	1000	0.1 [-0.2, 0.5]	0.0 [0.0, 0.0]	0.001 [0.001, 0.001]
2	0	0.50	Mean	1000	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	-0.002 [-0.002, -0.001]
2	0	0.50	Gaussian	1000	0.4 [0.0, 0.9]	0.0 [0.0, 0.0]	0.000 [0.000, 0.000]
2	0	0.50	Aug.	1000	0.1 [-0.2, 0.5]	0.0 [0.0, 0.0]	0.000 [0.000, 0.000]
2	1	0.25	Mean	1000	38.4 [35.0, 41.7]	0.0 [0.0, 0.0]	-1.317 [-1.338, -1.297]
2	1	0.25	Gaussian	1000	2.7 [0.5, 4.9]	0.0 [0.0, 0.0]	-0.102 [-0.104, -0.100]
2	1	0.25	Aug.	1000	2.0 [0.2, 3.9]	0.0 [0.0, 0.0]	-0.117 [-0.119, -0.115]
2	1	0.50	Mean	1000	49.9 [46.8, 53.2]	0.0 [0.0, 0.0]	-0.252 [-0.255, -0.249]
2	1	0.50	Gaussian	1000	0.2 [-1.8, 2.1]	0.0 [0.0, 0.0]	-0.065 [-0.066, -0.065]
2	1	0.50	Aug.	1000	0.9 [-0.9, 2.7]	0.0 [0.0, 0.0]	-0.067 [-0.068, -0.066]
12	0	0.25	Mean	1000	0.1 [0.0, 0.3]	0.0 [0.0, 0.0]	0.024 [0.021, 0.028]
12	0	0.25	Gaussian	1000	0.0 [-0.6, 0.6]	0.0 [0.0, 0.0]	-0.002 [-0.002, -0.001]
12	0	0.25	Aug.	1000	-0.3 [-0.7, 0.0]	0.0 [0.0, 0.0]	0.001 [0.000, 0.001]
12	0	0.50	Mean	1000	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	-0.002 [-0.002, -0.001]
12	0	0.50	Gaussian	1000	0.3 [-0.2, 0.8]	0.0 [0.0, 0.0]	0.000 [0.000, 0.000]
12	0	0.50	Aug.	1000	-0.1 [-0.4, 0.2]	0.0 [0.0, 0.0]	0.000 [0.000, 0.000]
12	1	0.25	Mean	1000	41.5 [38.2, 44.8]	0.0 [0.0, 0.0]	-1.202 [-1.219, -1.185]
12	1	0.25	Gaussian	1000	2.3 [0.2, 4.4]	0.0 [0.0, 0.0]	-0.104 [-0.106, -0.102]
12	1	0.25	Aug.	1000	2.3 [0.5, 4.1]	0.0 [0.0, 0.0]	-0.113 [-0.115, -0.111]
12	1	0.50	Mean	1000	50.8 [47.7, 54.1]	0.0 [0.0, 0.0]	-0.233 [-0.236, -0.230]
12	1	0.50	Gaussian	1000	0.1 [-1.8, 2.0]	0.0 [0.0, 0.0]	-0.065 [-0.066, -0.064]
12	1	0.50	Aug.	1000	1.2 [-0.6, 3.0]	0.0 [0.0, 0.0]	-0.064 [-0.065, -0.063]

Appendix S4. Household-resampled panel results

Each of 200 panel IDs supplied a prefix panel of $H = 1,200$ households and a panel of $H = 4,800$ households. Ten masks were retained within each panel at $p = 0.25$ and 0.50. Thus each method row below has 2,000 planned evaluations. Evaluations without a certified finite full-vector reference remain in the planned return denominator. Joint

coverage uses only certified-reference panels and counts method nonreturn there as zero. The Monte Carlo unit is the panel ID.

S4.1. Full-vector certification and extended focal references

The full-vector rule certifies 174 of the 200 small panels and all 200 large panels. The other 26 small panels all have original design rank 26, reduced rank 24 or 25, and a focal contrast passing the cone and row-space checks. The largest relative row-space residual is 1.52×10^{-15} . Their status is therefore `full_vector_target_not_regular`, rather than a certificate of focal nonidentification.

After the original analysis, a diagnostic extension reconstructed these 26 complete arrays from their archived donor addresses and fitted the reduced likelihood to *all* records. Every donor-index hash matched its archive. Both prescribed starts certified a finite focal coefficient in every panel, ranging from -0.4562 to 0.1511 . The maximum focal difference between starts was 2.10×10^{-13} ; the largest score per retained row and Newton correction were 1.85×10^{-13} and 1.92×10^{-10} . These fits verify the distinction between the full vector and focal contrast. They do not change the original target definition or reassign coverage in the original tables. No masked-data estimator was refitted for this check.

For Augmented, reference-conditional return is $1733/1740 = 99.60\%$ at $p = 0.25$ and $1740/1740 = 100\%$ at $p = 0.50$. Extended CC returns in all 1,740 certified-reference evaluations at either probability. The 4,000 panel-bootstrap draws give $[99.19\%, 99.94\%]$ for the former Augmented rate; bootstrap rates are constant in the all-return cells. These are empirical Monte Carlo intervals, not assurances of future return. The paired length comparison uses all pairs with two returned intervals: 1,734 and 1,740 pairs in the small panels. At $p = 0.25$, one of those pairs belongs to a panel without the original reference; its length is observable even though its original-reference coverage is undefined.

S4.2. Original performance tables

Table S4.1.: All household-resampled CPS panel cells. Each row has 2,000 planned evaluations. Valid counts certified full-vector references. Return uses all evaluations; coverage uses certified-reference evaluations. The final two columns require the reference and all ten intervals. Rates are percentages.

<i>H</i>	<i>p</i>	Method	Valid	Ret.	Joint	Cond.	Bias	RMSE	Length	Balanced	SE ² / <i>V_T</i>
1,200	0.25	CC	1740	3.9	4.3	94.9	-0.008	0.233	1.000	0	—
1,200	0.25	Mean	1740	87.0	87.0	87.0	-0.071	0.350	1.018	174	0.618
1,200	0.25	Ext. CC	1740	100.0	90.7	90.7	-0.063	0.391	1.147	174	0.636
1,200	0.25	Gaussian	1740	87.0	85.3	85.3	-0.006	0.207	0.606	174	0.592
1,200	0.25	Aug.	1740	86.7	95.9	96.3	-0.036	0.445	1.641	169	3.441
1,200	0.50	CC	1740	37.6	40.1	92.9	-0.006	0.157	0.542	1	—
1,200	0.50	Mean	1740	87.0	89.1	89.1	-0.056	0.231	0.696	174	0.661
1,200	0.50	Ext. CC	1740	100.0	92.1	92.1	-0.016	0.168	0.558	174	0.767
1,200	0.50	Gaussian	1740	87.0	90.8	90.8	-0.008	0.143	0.467	174	0.722
1,200	0.50	Aug.	1740	87.0	93.7	93.7	-0.009	0.164	0.592	174	0.897
4,800	0.25	CC	2000	84.6	80.4	95.0	-0.004	0.116	0.445	53	1.027
4,800	0.25	Mean	2000	100.0	86.8	86.8	-0.133	0.294	0.895	200	0.790
4,800	0.25	Ext. CC	2000	100.0	94.8	94.8	-0.006	0.118	0.446	200	0.966
4,800	0.25	Gaussian	2000	100.0	93.1	93.1	-0.012	0.113	0.406	200	0.874
4,800	0.25	Aug.	2000	100.0	94.7	94.7	-0.002	0.111	0.432	200	1.008

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Table S4.1 (continued)

H	p	Method	Valid	Ret.	Joint	Cond.	Bias	RMSE	Length	Balanced	SE^2/V_T
4,800	0.50	CC	2000	98.6	92.8	94.1	-0.001	0.066	0.246	180	0.912
4,800	0.50	Mean	2000	100.0	83.7	83.7	-0.065	0.123	0.361	200	0.796
4,800	0.50	Ext. CC	2000	100.0	94.1	94.1	-0.001	0.066	0.246	200	0.920
4,800	0.50	Gaussian	2000	100.0	93.4	93.4	-0.008	0.063	0.234	200	0.918
4,800	0.50	Aug.	2000	100.0	94.2	94.2	0.000	0.060	0.231	200	0.972

Table S4.2.: Paired Augmented minus Extended CC comparisons in the household-resampled panels. Brackets are 95% panel-bootstrap intervals, retaining all ten masks from a selected panel.

H	p	Planned	Valid	N_{both}	Δ Joint [MC]	Δ Ret. [MC]	Δ Length [MC]
1,200	0.25	2000	1740	1734	5.2 [3.7, 6.8]	-13.3 [-18.1, -8.9]	0.496 [0.400, 0.639]
1,200	0.50	2000	1740	1740	1.6 [0.3, 2.9]	-13.0 [-18.0, -8.5]	0.034 [0.027, 0.041]
4,800	0.25	2000	2000	2000	-0.1 [-1.2, 0.9]	0.0 [0.0, 0.0]	-0.014 [-0.017, -0.011]
4,800	0.50	2000	2000	2000	0.1 [-0.7, 1.0]	0.0 [0.0, 0.0]	-0.015 [-0.016, -0.014]

Table S4.3.: Non-available statuses in the household-resampled panels. Available counts are reported as interval-return counts in Table S4.1.

H	p	Method	Status	Count
1,200	0.25	CC	no_valid_root	1922
1,200	0.25	Mean	no_valid_root	260
1,200	0.25	Gaussian	no_valid_root	260
1,200	0.25	Aug.	ambiguous_roots	2
1,200	0.25	Aug.	no_valid_root	264
1,200	0.50	CC	no_valid_root	1249
1,200	0.50	Mean	no_valid_root	260
1,200	0.50	Gaussian	no_valid_root	260
1,200	0.50	Aug.	no_valid_root	260
4,800	0.25	CC	no_valid_root	308
4,800	0.50	CC	no_valid_root	28

Appendix S5. Protocol, computation and provenance

S5.1. Selected records and design matrices

The selection uses positive PULINENO, PRPERTYP= 2, PRPAGE $\geq$ 16, PEMLR $\in$ {1, 2}, PEIO1COW $\in$ {1, ..., 5}, HRMIS $\in$ {4, 8}, PEMJOT $\in$ {1, 2}, PXMJOT= 0, PTERNWA $\geq$ 0 and PRWERNAL= 0. We set $y = 1\{\text{PEMJOT} = 1\}$ and $z = \log(1 + \text{PTERNWA}/100)$. The 2025 and 2026 public-use layouts define the fields (U.S. Census Bureau, 2025, 2026). The January 2026 file is the archived basic-release file, not a substituted revision-archive release. Main-job earnings and usual hours describe different time concepts from the reference-week multiple-jobholding outcome (U.S. Bureau of Labor Statistics, 2026).

Let $a = \text{PRPAGE}$, $t = \text{PEHRUSL1}$, $g = 1\{1 \leq t \leq 99\}$, $t^* = gt$, $d = \text{PEEDUCA}$ and $(b)_+ = \max(b, 0)$. The 24 raw adjustment columns are:

Group	Columns	Count
Age	$a, (a - 25)_+, (a - 35)_+, (a - 45)_+, (a - 55)_+, (a - 65)_+$	6
Hours	$g, t^*, (t^* - 20)_+, (t^* - 35)_+, (t^* - 40)_+, (t^* - 50)_+$	6
Sex, marital status	$1\{\text{PESEX} = 2\}, 1\{\text{PEMARITL} = 1\}$	2
Education	$d, 1\{d \geq 39\}, 1\{d \geq 43\}$	3
Race	Indicators for $\text{PTDTRACE} = 2, = 4$, and outside $\{1, 2, 4\}$	3
Worker class	Indicators for $\text{PEIO1COW} = 2, 3, 4, 5$	4

The hours rule encodes responses outside 1–99 separately; these include the “hours vary” code and valid zero-hour responses, not only errors. Each adjustment column is centered and divided by its complete-array population standard deviation, bounded below by 10^{-8} . If $X_{\text{std}} = QR$ is the reduced QR factorization, the analysis uses $X = \sqrt{N}Q$. This transformation is fixed before masking and leaves the earnings coefficient unchanged when the raw design has full rank. Raw ranks are checked before relying on QR coordinates.

The richer learner matrix starts with the same age and hours terms and $\min(\text{HRNUMHOU}, 15)$, then categorical indicators for `PEMARITL`, `PESEX`, `PEEDUCA`, `PTDTRACE`, `HETENURE`, `GESTFIPS`, `GTCSASZ`, `PEIO1COW`, `PRCOW1`, `PRMJIND1` and `PRMJOC1`. Observed levels are sorted and the first is the reference; missing categories receive code -999 . Columns with standard deviation at most 10^{-12} are omitted before centering and scaling. An intercept, the outcome and reference-coded `PUSLFPRX` response-mode indicators are then appended; these appended columns are not standardized. The full learner dimensions are 140, 139 and 140 in chronological order. No earnings value is used to select these always-observed columns.

Households are keyed by the raw `HRHHID` and `HRHHID2` fields, joined by a vertical bar and sorted for integer indexing. CPS outer folds are the first 16 hexadecimal digits of the SHA-256 of `2026081902|household-key`, interpreted as an integer modulo 5. Repeated donor households in a panel receive distinct copy keys before fold assignment. Complete-array centering, columns and folds remain fixed throughout that array’s mask evaluations.

S5.2. Cross-fitted conditional mean and variance

Let b_i be a learner row as a column vector. For evaluation fold k , let $\mathcal{T}_k$ contain visible rows outside that household fold. With $P = \text{diag}(0, 1, \dots, 1)$, the fitted coefficients are

$$\hat{\eta}_k = \left(\sum_{i \in \mathcal{T}_k} b_i b_i^\top + \lambda_k P \right)^{-1} \sum_{i \in \mathcal{T}_k} b_i z_i, \quad \hat{\mu}_i = b_i^\top \hat{\eta}_k \quad (i \text{ in fold } k).$$

The penalty is on the sum of squared residuals, with an unpenalized intercept. The grid is $10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10$. Five inner household folds use the first 12 hexadecimal SHA-256 digits of `inner|k|h`, modulo 5, with zero-based integer household ID h . Validation squared errors are pooled over eligible inner folds; a fold requires at least 100 training and 10 validation rows. The first minimum in ascending grid order is selected. If no inner fold is eligible the implemented grid rule selects its first value. An outer training set with fewer than 200 visible rows causes nuisance failure.

Write $e_i = z_i - b_i^\top \hat{\eta}_k$, $n_k = |\mathcal{T}_k|$, $n_{kh} = |h \cap \mathcal{T}_k|$, $C_{kh} = n_{kh}(n_{kh} - 1)/2$ and $Q_{kh} =$

$\{(\sum_{i \in h \cap \mathcal{T}_k} e_i)^2 - \sum_{i \in h \cap \mathcal{T}_k} e_i^2\}/2$. With $\epsilon = 10^{-6}$,

$$\begin{aligned} t_k^0 &= n_k^{-1} \sum_{i \in \mathcal{T}_k} e_i^2, & t_k &= \max(t_k^0, \epsilon), \\ D_k &= \sum_h C_{kh}, & \tau_k^0 &= \frac{\sum_h Q_{kh}}{\max(D_k, 1)}, \\ \tau_k &= \max(\tau_k^0, \epsilon), & v_k &= \tau_k + \max(t_k - \tau_k, \epsilon). \end{aligned}$$

Every evaluation row in fold k uses $\hat{v}_i = v_k$. The Gaussian integration uses 30 physicists' Hermite nodes and their weights divided by $\sqrt{\pi}$, with node values $\hat{\mu}_i + \sqrt{2v_k}\xi_l$.

S5.3. Household slopes for the unpenalized procedures

For each fitted score sum Ψ_j , define $A_j = -N^{-1}\partial\Psi_j/\partial\theta^\top$ with the learner held fixed and solve $A_j^\top \ell_j = c$, where c selects the earnings coefficient. For CC and Augmented the slopes used in the variance are

$$q_{CC,h} = \frac{1}{N} \ell_{CC}^\top \sum_{i \in h} s_i, \quad q_{A,h} = \frac{1}{Np} \ell_A^\top \sum_{i \in h} (s_i - t_{Gi}), \quad R_h = 1.$$

All functions are evaluated at the corresponding fitted coefficient. The Augmented calculation uses the observed-minus-generated score; it does not add a fitted-learner derivative term.

For Mean and Gaussian, keep each selected λ_k fixed and perturb the weight of an observed training household h . With $S_k = \sum_{i \in \mathcal{T}_k} b_i b_i^\top + \lambda_k P$, its ridge slope is

$$\dot{\eta}_{kh} = S_k^{-1} \sum_{i \in h \cap \mathcal{T}_k} b_i e_i.$$

Define $E_k = \sum_{i \in \mathcal{T}_k} b_i e_i$ and $F_k = \sum_{i \in \mathcal{T}_k} b_i (\sum_{a \in h(i) \cap \mathcal{T}_k} e_a - e_i)$. The residual-moment slopes are

$$\begin{aligned} \dot{t}_{kh}^0 &= \frac{\sum_{i \in h \cap \mathcal{T}_k} e_i^2 - n_{kh} t_k^0 - 2E_k^\top \dot{\eta}_{kh}}{n_k}, \\ \dot{\tau}_{kh}^0 &= \frac{Q_{kh} - 1\{D_k > 1\} \tau_k^0 C_{kh} - F_k^\top \dot{\eta}_{kh}}{\max(D_k, 1)}, \\ \dot{t}_{kh} &= 1\{t_k^0 > \epsilon\} \dot{t}_{kh}^0, \quad \dot{\tau}_{kh} = 1\{\tau_k^0 > \epsilon\} \dot{\tau}_{kh}^0, \\ \dot{v}_{kh} &= \begin{cases} \dot{t}_{kh}, & t_k - \tau_k > \epsilon, \\ \dot{\tau}_{kh}, & t_k - \tau_k < \epsilon. \end{cases} \end{aligned}$$

Pair sums receive one weight per household. In the first branch the pair covariance cancels from $v_k = t_k$. The effective branch must be separated from its boundary under the checks below.

Let $f_{ji} = \ell_j^\top t_{ji}$, $f_{ji,\mu} = \partial f_{ji} / \partial \mu_i$ and $f_{ji,v} = \partial f_{ji} / \partial v_i$. The implemented slope is

$$q_{jh} = \frac{1}{N} \left[\ell_j^\top \sum_{i \in h} (s_i - t_{ji}) + (1-p) \sum_k \sum_{i: \text{fold}(i)=k} \{f_{ji,\mu} b_i^\top \dot{\eta}_{kh} + f_{ji,v} \dot{v}_{kh}\} \right], \quad j = M, G. \quad (\text{S5.1})$$

Here a training-household slope is zero for its own evaluation fold. The factor $1-p$ is the expected missingness weight in the expansion; it is not replaced by realized $1-R_i$ in the reported procedure. The derivative of the discrete CV argmin is not taken, so the approximation requires stability of the selected branch.

These generated-score derivatives are explicit. At $w_i(z) = (1, z, x_i^\top)^\top$ and $\pi_i(z) = \text{expit}\{w_i(z)^\top \hat{\theta}_j\}$, put

$$d_{ji}(z) = \ell_{j,2} \{y_i - \pi_i(z)\} - \hat{\beta}_j \pi_i(z) \{1 - \pi_i(z)\} \ell_j^\top w_i(z).$$

For Mean, $f_{Mi,\mu} = d_{Mi}(\hat{\mu}_i)$ and $f_{Mi,v} = 0$. For Gaussian, with $z_{il} = \hat{\mu}_i + \sqrt{2v_k} \xi_l$,

$$f_{Gi,\mu} = \sum_l \omega_l d_{Gi}(z_{il}), \quad f_{Gi,v} = \frac{1}{\sqrt{2v_k}} \sum_l \omega_l \xi_l d_{Gi}(z_{il}).$$

These are derivatives of the implemented quadrature, not derivatives of an uncomputed exact integral.

For Extended CC, let D_i be a row of the retained, scaled design in an orthonormal basis for its row space. If a represents the original earnings contrast in those coordinates and $J = \sum_{i \in \mathcal{F}_R} D_i D_i^\top \hat{\pi}_i (1 - \hat{\pi}_i)$, then

$$q_{E,h} = a^\top J^{-1} \sum_{i \in h \cap \mathcal{F}_R} D_i (y_i - \hat{\pi}_i).$$

The removed rows have zero limiting residual. This observed-face plug-in calculation has the restricted justification in Section S1; arbitrary face selection is not covered by that proposition. Every procedure uses $\hat{s}_j^2 = H(H-1)^{-1}(1-p) \sum_{h: R_h=1} q_{jh}^2$. The N^{-1} factor is already in the full-vector slopes above; the reduced formula instead uses a total, rather than averaged, information matrix.

S5.4. Firth penalty and its mask derivative

Firth maximizes the observed sum log likelihood plus $\frac{1}{2} \log \det I$, without scaling the likelihood by $1/p$. To state the derivatives, write w_i for the full logistic design and $r_i = r_{h(i)}$ for a continuous household weight. At the observed fit let $\pi_i = \text{expit}(w_i^\top \theta)$, $v_i = \pi_i(1 - \pi_i)$, $d_i = 1 - 2\pi_i$, $I = \sum_i r_i v_i w_i w_i^\top$ and $t_i = w_i^\top I^{-1} w_i$. If $I_a = \sum_i r_i v_i d_i w_{ia} w_{i\alpha}^\top$, the adjusted score and negative Jacobian are

$$U = \sum_i w_i \{r_i (y_i - \pi_i) + \frac{1}{2} r_i v_i t_i d_i\},$$

$$J = I - \frac{1}{2} \sum_i r_i v_i t_i (d_i^2 - 2v_i) w_i w_i^\top + \frac{1}{2} [\text{tr}(I^{-1} I_a I^{-1} I_b)]_{ab}.$$

The derivative with respect to an individual weight has component

$$d_{ia}^F = w_{ia}(y_i - \pi_i) + \frac{1}{2}v_i d_i t_i w_{ia} - \frac{1}{2}v_i w_i^\top I^{-1} I_a I^{-1} w_i.$$

Thus $q_{F,h} = c^\top J^{-1} \sum_{i \in h} d_i^F$ enters the same household variance formula. The final term includes changes to other rows' leverages through I . Omitting it would not differentiate the Jeffreys penalty with respect to the mask weights. The implementation uses an invertible SVD change of coordinates and transforms the earnings contrast accordingly.

At common full-array weight p , the penalized center solves $pS_F(\theta) + A_F(\theta) = 0$, where A_F is the penalty score. It generally differs from the unpenalized target. We retain that difference in evaluating coverage and bias. The resulting normal interval is an approximate household-mask interval, not the usual Firth profile-likelihood interval. Finite point estimates under the conditions of Kosmidis and Firth (2021) do not establish coverage under random loss of household support.

S5.5. Masks, pairing and Monte Carlo units

For the fixed monthly arrays, a household was observed with known probability p . January 2025 at $p = 0.25$ and 0.50 reused 6,400 archived mask addresses; the other seven month- p cells used 2,500 masks each, for 30,300 mask addresses and 181,800 method rows. The new addresses used NumPy `SeedSequence` with master components `[20260918, 2, 406183, phase, YYYYMM, rep, component]`. Archived masks were accepted only when their packed-bit SHA-256 values matched. A 4,000-draw bootstrap resampled mask IDs within a month- p cell and shared each draw across methods and paired contrasts. Failed masks stayed in the draw. New mask addresses use phase 1, component 0 and repetitions 0–2,499; their common household uniforms are thresholded at each p . The two older January cells use `[20260916, 4, 731863, int(100*p), 0, rep]`, with repetitions 0–6,399; these older masks are not nested across p . The fixed-array bootstrap uses phase 2, repetition 0 and component $100p$. All streams use NumPy's `default_rng` with PCG64. No pooled test treats the monthly survey arrays as independent population samples.

For C6, the B^+ and B procedures shared repetition ID, complete data, target and mask. The 4,000 paired draws resampled the 1,000 repetition IDs from `SeedSequence([20260918, 2, 406183, 6, 0])` and were shared across conditions and methods. For the empirical panels, donor households were sampled at phase 3; repeated donors received distinct cluster IDs. A single length-4,800 donor sequence supplied both panel sizes, and one length-4,800 uniform sequence for each of ten phase-4 mask IDs supplied both H values and both observation probabilities. Phase 5 resampled 200 panel IDs 4,000 times, retaining the ten masks within a selected panel. No panel or mask was redrawn after target or fitting failure.

S5.6. Numerical certification and failure rules

CC, Mean, Gaussian and Augmented solve averaged scores, starting from zero and from an intercept-only vector with intercept $\text{logit}\{(\sum_i y_i + 0.5)/(N + 1)\}$ based on always-observed outcomes. Newton iteration allows 60 steps, maximum coordinate step 10, 24 line-search halvings and Armijo constant 10^{-4} on the squared-score merit. Internal score and correction thresholds are 10^{-10} and 10^{-8} . If the final candidate is

uncertified, the prespecified trust-region least-squares search on the same score uses at most 300 function evaluations and tolerances 10^{-12} . This search changes no estimating equation and its software success flag is not a root certificate.

The final certificate requires averaged score norm at most 2×10^{-8} , Newton correction at most 2×10^{-6} , linear backward error at most 10^{-10} , and full numerical rank of the Jacobian. Rank uses singular values greater than $\epsilon_{\text{mach}} d \sigma_{\text{max}}$ in the original coefficient coordinates. A linear solve $Ax = b$ has backward error $\|Ax - b\|_{\infty} / (\|A\|_{\infty} \|x\|_{\infty} + \|b\|_{\infty})$. When both starts certify, all coordinate differences must be at most $2 \times 10^{-5} \{1 + \max(|\theta_{1a}|, |\theta_{2a}|)\}$. One certified start is accepted as **single_start_valid**; two disagreeing certified roots give **ambiguous_roots**; no certified start gives **no_valid_root**. Thus two starts are tried, but both need not succeed for these four procedures.

Extended CC scales each design column by its maximum absolute value, bounded below by 1 only when that maximum is zero. Its cone programs constrain each direction coordinate to $[-1, 1]$, use primal and dual feasibility tolerances 10^{-9} , and require primal residual, dual residual and duality gap at most 2×10^{-7} . Facial reduction stops when the mean remaining signed margin is at most 10^{-9} ; margins above 10^{-7} identify rows to remove. Ambiguous strict margins or more than $d + 2$ rounds cause numerical failure. The focal cone extent and relative row-space residual must both be at most 10^{-8} . Reduced-design rank uses $\max(n, d) \epsilon_{\text{mach}} \sigma_{\text{max}}$. Both reduced-likelihood starts must meet score-per-row 2×10^{-12} and Newton correction 2×10^{-10} , within 100 steps and 50 line-search halvings. Focal coefficients must agree within $10^{-8} \max(1, |\hat{\beta}|)$. The starts are zero and the least-squares representation of the constant smoothed outcome logit on the retained design.

Firth requires full observed-design rank at relative tolerance 10^{-12} . Both SVD-coordinate starts must have score per row at most 10^{-9} , Newton correction at most 2×10^{-7} , positive curvature and curvature condition number below 10^{12} . The transformed-back coefficients must agree at absolute-plus-relative tolerance 2×10^{-6} . Its search allows 150 steps, coordinate steps at most 5, and 30 halvings; internal score and correction tolerances are 10^{-11} and 10^{-9} . A Fisher direction replaces a nonascending Newton direction, within the same prespecified penalized objective. These checks certify the reported stationary fit, not uniqueness of every possible penalized maximum.

CC, Mean, Gaussian and Augmented focal loading solves require backward error at most 10^{-10} . For the learner-aware variance, reconstructed fold means and variances must agree with the fitted learner within 10^{-6} and 10^{-8} , and the ridge normal-equation relative residual must be at most 10^{-10} . Gaussian intervals are withheld within 10^{-10} of a relevant boundary of the effective variance-floor map, including the pair-count branch when it affects that map. A floor in τ that cancels from $v = t$ does not by itself trigger failure. The selected CV branch is held fixed; these checks do not account for uncertainty from switching penalties.

Intervals require a certified point and finite positive SE. A failed learner is recorded as **nuisance_failure**; uncertified roots and start disagreement retain the statuses above. Geometry can produce **focal_not_estimable**, and failed SE construction produces **variance_failure**. Empty observed samples receive **no_observed_rows**. Firth additionally distinguishes **observed_design_rank_deficient**, **uncertified_stationary_point** and **starts_disagree**. Numerical exceptions have a **numerical_failure** status and a reason. No new solver, penalty or adjustment-column deletion is introduced after examining a failure. The fixed reduced-basis operation in Extended CC is part of that procedure's definition.

S5.7. Reporting sets and nested variance

Every planned unit is retained for overall point and interval return. Reference-conditional return and joint coverage use only panels with the prespecified finite full-vector reference; conditional coverage further requires interval return. Bias and RMSE require that reference and a returned point. Per-method panel length and SE summaries require the reference and a returned interval, while paired length differences require both intervals and do not require a reference value. Each such pairing count is reported.

For the nested diagnostic, an eligible panel must have a certified reference and all ten point estimates and intervals returned. For eligible panel b and mask k , let $e_{bk} = \hat{\beta}_{bk} - \beta_{F,b}$ and $\bar{e}_b = 10^{-1} \sum_k e_{bk}$. If $L \geq 2$ panels qualify, define

$$W = \frac{1}{L} \sum_b \frac{1}{9} \sum_k (e_{bk} - \bar{e}_b)^2, \quad B = \frac{1}{L-1} \sum_b (\bar{e}_b - \bar{e})^2.$$

The adjusted between-panel component is $B - W/10$ and total error variance is $V_T = B + 0.9W$. Mean reported SE squared is averaged over the same $10L$ intervals. These components describe errors about each panel's own reference, conditional on complete return. They do not estimate a survey-design population variance. Negative adjusted components are retained. If fewer than two panels qualify, all nested variance quantities are undefined.

S5.8. Reproducibility and checks

The replication record contains the fitting code, raw per-fit outputs, source hashes and random addresses. The companion `REPLICATION_GUIDE.md` maps those files to the analysis stages and gives the public CPS download sources. The complete monthly design matrices, category levels and data-version hashes are recorded separately from the masked fitting inputs. Source versions are held fixed rather than silently replaced by later downloads.

Independent archive checks replayed the original masks and reconstructed the fixed-array and panel summaries from their raw records. The additional 26-panel complete-data focal fits and reference-conditional return intervals are explicitly identified as revision diagnostics in `revision_20260919`; they leave the original score fits, masks, reference rule and performance tables unchanged. Operational development history and full hash manifests are retained with the replication files, while the scientific definitions and numerical rules are given above. Internal reproducibility does not convert the experimental mask law into actual CPS nonresponse or a population sampling design.

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